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Steady-state modelling of hybrid energy system for off grid electrification of cluster of villages

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ABSTRACT

Electrification of villages from the main grid leads to large investments and losses, and this forms the basis of decentralized Hybrid Energy System. In order to evaluate the techno-economic performance of hybrid energy system for remote rural area electrification, a mixed integer linear mathematical programming model (time-series) has been developed to determine the optimal operation, optimal configuration including the assessment of the economic penetration levels of photovoltaic array area, and cost optimization for a hybrid energy generation system consisting of small/micro hydro based power generation, biogas based power generation, biomass (fuelwood) based power generation, photovoltaic array, a battery bank and a fossil fuel generator. An optimum control algorithm written in C++, based on combined dispatch strategy, allowing easy handling of the models and data of hybrid energy system components is presented. A special feature of the proposed model is that a cost constant (cost/unit) for each of the proposed resource is introduced in the cost objective function in such a way that resources with lesser unit cost share the greater of the total energy demand in an attempt to optimize the objective function.

To demonstrate the use of model and algorithm, a case study for a rural remote area is also presented.
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1. Introduction

Hybrid Energy Systems (HES) generally integrate renewable energy sources with fossil fuel powered diesel/petrol generator to provide electric power where the electricity is either fed directly into the grid or to batteries for energy storage. The role of integrating renewable energy in a hybrid energy system is primarily to save diesel fuel.

A hybrid energy system may or may not be connected to the grid. They are generally independent of large centralized electric grids and are used in rural remote areas. In these systems it is possible for the individual power sources to provide different percentages of the total load.

For systems employing totally clean renewable energy, high capital cost is an important barrier. However, we can produce green power by adding different renewable energy sources to diesel generator and battery, which is also called a hybrid energy system. This kind of system can compromise investment cost, diesel fuel usage cost and also operation and maintenance costs.

The widespread utility acceptance of hybrid energy system has left something to be desired. The most important aspects and also main concern of hybrid energy system is to supply power to the customers economically. The economics of hybrid energy system depends on two factors: the size of the individual component and the dispatch strategy [1].

In view of those factors, three logical and simple questions arise: (i) How power supplies are shared among generators in a hybrid energy system in the most economic manner? (ii) What should be the economic operation control strategy for determining power flow among different generation units?

In practical decentralized hybrid energy systems, there are often different renewable generators, diesel generators and battery storage. An over sizing of the photovoltaic array area significantly elevates the overall cost of the hybrid energy system. In view of high system cost of photovoltaic (PV), it would be judicious to minimize the penetration of PV [6]. That's why the third and final question is (iii) What is the economic optimal proportion (penetration) of PV array area into a hybrid energy system [7]?

The economic penetration level is that which achieves the maximum permissible reduction in the kWh cost of hybrid energy system [8].

Hybrid energy system economics and studies related to the assessment of the economic penetration level of photovoltaic have

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List of symbols

E_j Hourly energy output from generating/discharge unit j, kWh
j j stands for MHG, BGG, BMG, PVG, DEG, and BATT
$E_{j,\text{Load}}$ Energy output from generating/discharge unit j directed to load
$E_{j,\text{BATT}}$ Energy output from generating unit j directed to battery
$E_{j,\text{Dump}}$ Excess energy from generating unit j directed to dump load
$E_{\text{BATT,IN}}$ Energy input to battery
Q Discharge in m^3/s
ρ Density of water = 1000 kg/m^3
h Head in m
e Effective power delivery factor
η_{MHG} Overall efficiency of micro-hydro generator
η_{DFEG} Efficiency of dual fuel engine generator
$G(t)$ Hourly irradiance in kWh/m^2
A Surface area of the PV modules in m^2
P PV penetration level factor
η_{PVG} Efficiency of PV generator
η_{DEG} Diesel generator efficiency
$E_{\text{REC-OUT}}(t)$ Hourly energy output from rectifier, kWh
$E_{\text{REC-IN}}(t)$ Hourly energy input to rectifier, kWh
$E_{\text{SUR-AC}}(t)$ Amount of surplus energy from AC sources, kWh
η_{REC} Efficiency of rectifier
$E_{\text{SUR-DC}}(t)$ Amount of surplus energy from DC sources, kWh
$E_{\text{CC-OUT}}(t)$ Hourly energy output from charge controller, kWh
$E_{\text{CC-IN}}(t)$ Hourly energy input to charge controller, kWh
η_{CC} Efficiency of charge controller
$E_{\text{PVG-INV}}(t)$ Hourly energy output from inverter (in case of SPV), kWh
$E_{\text{BATT-INV}}(t)$ Hourly energy output from inverter (in case of battery), kWh
η_{INV} Efficiency of inverter
$E_{\text{Needed}}(t)$ Hourly energy needed by the load side, kWh
$E_{\text{BATT}}(t)$ Energy stored in battery at hour t, kWh
$E_{\text{BATT}}(t-1)$ Previously stored energy in battery, kWh
η_{CHG} Battery charging efficiency
η_{DCHG} Battery discharging efficiency
E_{BATmax} Maximum capacity of battery, kWh
$E_{\text{Load}}(t)$ Total hourly load at time t, kWh
SOC_{min} Minimum state of charge of battery
SOC_{max} Maximum state of charge of battery
E_{BATTmin} Minimum energy stored in battery, kWh
E_{BATTmax} Maximum energy stored in battery, kWh
CV_{BG} Calorific value of biogas = 4700 kcal
η_{CBG} Biogas conversion efficiency
CV_{BM} Calorific value of biomass (fuelwood) = 4015 kcal
η_{CBM} Biomass conversion efficiency
H Total daily potential of design month, kWh/m^2
PVG Direct Load PVG energy delivered directly to the load, kWh
Battery Load Battery energy delivered directly to the load, kWh
$\text{TDE}_{\text{Surplus}}$ Total daily surplus energy from renewable, kWh
$\text{TDE}_{\text{Netload}}$ Total daily netload demand, kWh
P_{BATT} Battery capacity, kWh
K Maximum permissible time to operate the generator daily
T_{DEG} Number of running hours daily.

been presented in the literature [1–5] and [6–8] respectively. In [2], a mathematical model has been introduced to determine the optimal configuration of the hybrid energy system while satisfying system operational constraints. Model is solved by HyperLindo software without using any dispatch strategy. In [3], a model has been introduced to estimate the optimized cost and solved by dispatch strategy based solution algorithm but does not based on any cost-effective optimization approach or energy balance concept. Also, the model developed in [2,3] does not include the determination of economic penetration level of photovoltaic array area.

In [7], a method has been introduced for the assessment of the optimal proportion (penetration) of PV. The method is based on evaluation of costs and loss of load risk (LOLR). In [8], an operation strategy has been developed on the basis of loss of load expectation (LOLE). Both the above mentioned method is not based on any cost optimization approach.

In this paper, model design approach in [2,3] and solution approach [3] has been extended to include the cost effective approach with hourly energy balance concept while introducing generalize mixed integer linear mathematical programming model and combined dispatch strategy based solution algorithm, based on the evaluation of optimized system unit cost, to determine the optimal operation, optimal sizing including the assessment of the economic penetration levels of photovoltaic array area, and cost optimization for a hybrid energy system. For the determination of economic penetration level, a PV penetration level factor (P) is also incorporated in the model.

In order to consider the fluctuation in the discharge and power generation from least unit cost stand-alone micro-hydro generation system, a sensitivity factor, EPDF (e) has also been included that brings to completion all the operational aspects of the hybrid energy system that have an impact on the reduction of operational costs. The EPDF is Electric Power Delivery Factor and also called optimizing power factor and is maximum equal to 1.

A cost effective approach of the proposed model is that a cost constant (cost/unit) for each of the proposed resource is introduced in the cost optimization function in such a manner that resources with lesser unit cost share the greater of the total energy demand in an attempt to optimize the objective function.

This paper is organized as follows: in the next section the system configuration is given. Sections 3 and 4 present the system component modelling and the development of model respectively. Section 5 presents the proposed combined dispatch strategy. Section 6 presents the system control. In Section 7, the economic dispatch algorithm is presented. The case study and simulation results are given in Section 8. Section 9 concludes our study.

2. Hybrid energy system configuration

The block diagram for a typical stand-alone hybrid energy system, based on a generalized three-bus configuration is shown in Fig. 1. The system consists of micro-hydro generator (MHG), biogas generator (BGG), biomass (fuelwood) generator (BMG), photovoltaic generator (PVG), battery bank storage, back up diesel generator (DEG), and dump load. Provisions for the availability of both AC and DC buses are made using electronic converters.

To serve the load, electrical energy can be produced either directly from renewable generators and diesel generator, or indirectly from the battery bank. These relationships are expressed in Eqs. (1.1–1.5).

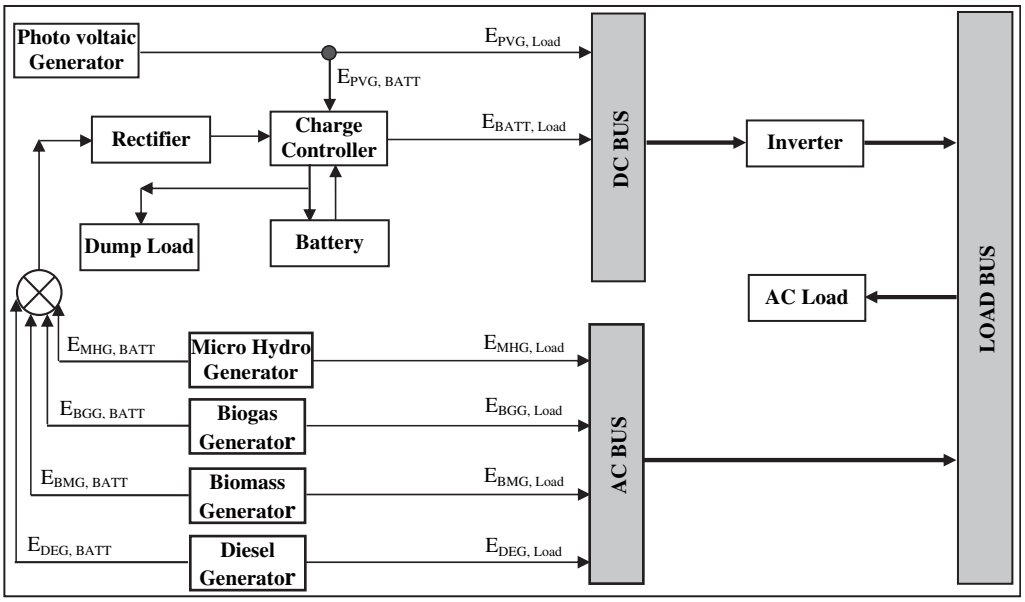


Fig. 1. The configuration of hybrid energy system.

$$E_{PVG}(t) = E_{PVG,Load}(t) + E_{PVG,BATT}(t) + E_{PVG,Dump}(t) \quad (1.1)$$

$$E_{MHG}(t) = E_{MHG,Load}(t) + E_{MHG,BATT}(t) + E_{MHG,Dump}(t) \quad (1.2)$$

$$E_{BGG}(t) = E_{BGG,Load}(t) + E_{BGG,BATT}(t) + E_{BGG,Dump}(t) \quad (1.3)$$

$$E_{BMG}(t) = E_{BMG,Load}(t) + E_{BMG,BATT}(t) + E_{BMG,Dump}(t) \quad (1.4)$$

$$E_{DEG}(t) = E_{DEG,Load}(t) + E_{DEG,BATT}(t) + E_{DEG,Dump}(t) \quad (1.5)$$

In any hour t, the energy available to charge the battery bank (BATT) is shown in Eq. (1.6), and the energy available from the battery to serve the load is shown in Eq. (1.7).

$$E_{BATT,IN}(t) = \eta_{CHG} \times \eta_{CC} \times \left[E_{PVG,BATT}(t) + \eta_{REC} \times \left(E_{MHG,BATT}(t) + E_{BGG,BATT}(t) + E_{BMG,BATT}(t) + E_{DEG,BATT}(t)\right)\right] \quad (1.6)$$

$$E_{BATT,Load}(t) = \eta_{DCHG} \times \left[E_{BATT,IN}(t)\right] \quad (1.7)$$

Finally, the total energy available to serve the load is written in Eq. (1.8).

$$E_{Load}(t) = \left[E_{MHG,Load}(t) + E_{BGG,Load}(t) + E_{BMG,Load}(t) + E_{DEG,Load}(t) + \eta_{INV} \times \left(E_{PVG,Load}(t) + E_{BATT,Load}(t)\right)\right] \quad (1.8)$$

3. System component modelling

Modelization is an essential step before any phase of optimization. For a hybrid energy system with battery storage, six principal subsystems are included, MHG, BGG, BMG, PVG, DEG and the battery bank. The operating performances of these components play an important role in system applications and system evaluations. A methodology with general description for modelling hybrid energy system components is described below.

3.1. Mathematical model of micro hydro generator

The hydro power usually refers to the generation of shaft power from falling water. The power is then used for direct mechanical purposes or, more frequently, for generating electricity. The theoretical electrical power generated by the micro-hydro unit (P_{MHG}) is given by

$$P_{MHG}(t) = 9.81 \times Q \times \rho \times h \quad (2.1)$$

and the total energy (E_{MHG}) in kWh in an hour t is given by

$$E_{MHG}(t) = [P_{MHG}(t) \times e] \times \eta_{MHG} \quad (2.2)$$

3.2. Mathematical model of biomass generator

Producer gas is a low calorie mixture of carbon monoxide and methane with non combustible gases. It is obtained when fuelwood (biomass) is burned with insufficient oxygen (fuelwood gasification). Some special diesel engines that use producer gas and diesel in a certain ratio are available, and when coupled to an alternator, decentralized electricity can be produced.

The annual delivered electricity output (E_{Annual}) of a biomass gasifier energy system with rated power output (P_{BMG}) of electricity generator is dependent on its capacity utilization factor (CUF). Assuming 21% conversion efficiency from fuelwood to electricity, it can be modelled using the following expression:

$$E_{Annual} = P_{BMG}(8760 \times CUF) \quad (3.1)$$

and hourly energy output (E_{BMG}) is given by

$$E_{BMG}(t) = P_{BMG}(t) \times \eta_{DFEG} \quad (3.2)$$

3.3. Mathematical model of biogas generator

Decentralized electricity generation using biogas and diesel in the ratio of 80:20 in a diesel engine (dual fuel mode) coupled to an alternator. Biogas is produced from livestock manure and human sewage. Assuming 27% conversion efficiency from biogas to

electricity, it can be modelled using the following expression as explained above:

$$E_{\text{Annual}} = P_{\text{BGG}}(8760 \times \text{CUF}) \tag{4.1}$$

and hourly energy output (E_{BGG}) is given by

$$E_{\text{BGG}}(t) = P_{\text{BGG}}(t) \times \eta_{\text{DFEG}} \tag{4.2}$$

3.4. Mathematical model of SPV generator

Solar photovoltaic (SPV) technology involves the direct conversion of sunlight into electricity through the use of photovoltaic modules. The sunlight impinging on panels, i.e. insolation, is measured in units of W/m^2. The PV system power output (DC) has approximately a linear relationship to the insolation. Using the solar radiation available on the tilted surface the hourly energy output (E_{PVG}) of the PV generator, can be calculated according to the following equation [9]:

$$E_{\text{PVG}}(t) = G(t) \times A \times P \times \eta_{\text{PVG}} \tag{5.1}$$

All the energy losses in a PV generator, including connection losses, wiring losses and other losses, are assumed to be zero. Eq. (5.1) assumes that PV generator has a tracking system and a maximum power point tracker (i.e. $\eta = 1$). It also assumes that the temperature effects (on PV cells) are ignored.

3.5. Mathematical model of diesel generator

Conventional generators are normally diesel engines coupled to generator. Diesel generators supply energy in one of two ways. Either they generate only the power needed by the load (load following), or they generate at nominal power and the surplus energy (if any), is used to charge the battery bank (cycle charging). In this study, a diesel generator of both kinds is considered. The generator model is designed in such a way that the diesel generator is always operate between 80 and 100% of their kW rating, while operating in conjunction with the battery bank and other renewable generators. Hourly energy generated (E_{DEG}) by diesel generator with rated power output (P_{DEG}) is defined by the following expression:

$$E_{\text{DEG}}(t) = P_{\text{DEG}}(t) \times \eta_{\text{DEG}} \tag{6.1}$$

3.6. Mathematical model of rectifier

The rectifier is used to transform the surplus AC power from the MHG, BGG, BMG, and DEG to DC power of constant voltage (when the energy generated by the hybrid energy system exceeds the load demand). The rectifier model is given below:

$$E_{\text{REC-OUT}}(t) = E_{\text{REC-IN}}(t) \times \eta_{\text{REC}} \tag{7.1}$$

$$E_{\text{REC-IN}}(t) = E_{\text{SUR-AC}}(t) \tag{7.2}$$

at any time t,

$$E_{\text{SUR-AC}}(t) = E_{\text{MHG}}(t) + E_{\text{BGG}}(t) + E_{\text{BMG}}(t) + E_{\text{DEG}}(t) - E_{\text{Load}}(t) \tag{7.3}$$

3.7. Mathematical model of charge controller

To prevent overcharging of a battery, a charge controller is used to sense when the batteries are fully charged and to stop or decrease the amount of energy flowing from the energy source to the batteries. The model of the charge controller is presented below:

$$E_{\text{CC-OUT}}(t) = E_{\text{CC-IN}}(t) \times \eta_{\text{CC}} \tag{8.1}$$

$$E_{\text{CC-IN}}(t) = E_{\text{REC-OUT}}(t) + E_{\text{SUR-DC}}(t) \tag{8.2}$$

3.8. Mathematical model of inverter

The photovoltaic generator and battery produce DC power and therefore when the hybrid energy system contains an AC load, a DC/AC conversion is required. The inverter model for photovoltaic generator and battery bank are given below:

$$E_{\text{PVG-INV}}(t) = E_{\text{PVG}}(t) \times \eta_{\text{INV}} \tag{9.1}$$

$$E_{\text{BATT-INV}}(t) = [(E_{\text{BATT}}(t-1) - E_{\text{Load}}(t))/(\eta_{\text{INV}} \times \eta_{\text{DCHG}})] \tag{9.2}$$

3.9. Mathematical model of battery bank

The battery state of charge (SOC) is the cumulative sum of the daily charge/discharge transfers. At any hour t the state of battery is related to the previous state of charge and to the energy production and consumption situation of the system during the time from $t-1$ to t.

During the charging process, when the total output of all generators is greater than the load demand, the available battery bank capacity at hour t can be described by

$$E_{\text{BATT}}(t) = E_{\text{BATT}}(t-1) + E_{\text{CC-OUT}}(t) \times \eta_{\text{CHG}} \tag{10.1}$$

On the other hand, when the load demand is greater then the available energy generated, the battery bank is in discharging state. Therefore, the available battery bank capacity at hour t can be expressed as:

$$E_{\text{BATT}}(t) = E_{\text{BATT}}(t-1) - E_{\text{Needed}}(t) \tag{10.2}$$

$$E_{\text{Needed}}(t) = E_{\text{Netload}}(t)/(\eta_{\text{INV}} \times \eta_{\text{DCHG}}) \tag{10.3}$$

Meanwhile, the charged quantity of the battery is subject to the following constraints:

$$\text{SOC}_{\min} \le \text{SOC}(t) \le \text{SOC}_{\max} \tag{10.4}$$

The maximum value of SOC is 1, and the minimum SOC is determined by maximum depth of discharge (DOD),

$$\text{SOC}_{\min} = 1 - \text{DOD} \tag{10.5}$$

3.10. Mathematical model of dump load

The dump energy is defined as the energy produced by the renewable generators or diesel generator but unused when the load does not need all the energy and the battery has reached its maximum capacity and can not store more energy [9]. In this study, a conventional electric water heater is assumed as dump load. The dump energy for hour t is calculated as follows:

$$E_{\text{Dump}}(t) = E_{\text{CC-OUT}}(t) - [(E_{\text{BATMAX}} - E_{\text{BATT}}(t-1))/\eta_{\text{CHG}}] \tag{11.1}$$

Negative results are assumed as zero dump energy.

4. Development of model

This section describes an optimization technique developed for the design of a stand-alone hybrid energy system. The stand-alone hybrid energy system in this research serves the isolated demand in cluster of rural remote villages. The design takes into account the unit cost of different resources based on life cycle costs. The results are optimal suggestion for optimal configurations based on locally available energy resources. The problem is formulated as a mixed integer linear programming developed in C++ [2,3].

The mathematical approach is presented in a simple and useful form directly applicable for the modelling of stand-alone hybrid energy system for use in the remote rural areas. The input data required, assumptions involved, and the design procedure are clearly explained. The approach involves the minimization of a cost function subject to a set of equality and inequality constraints.

4.1. Model assumptions

In order to state a model which is both sufficiently general and accurate for describing all types of energy flow, we make following assumptions and simplifications:

- We consider the system in steady state.
- We consider the steady state power, efficiency, and energy only, no other values are used for the system description.
- The model incorporates conservation laws (e.g. conservation of flow) but no constitutional laws (e.g. relation between voltage and current).
- Since the electrification process is continuous, it is assumed that the hybrid generation system operates on a 24 h a day and therefore equipment breakdowns, planned maintenance etc., are not explicitly considered.
- It is assumed that the costs are additive and that there are no perceptional changes in other factors such as plant capacity, unit cost of different energy sources, configuration of energy system etc.
- It is assumed that the available technologies at the beginning of planning horizon will remain available throughout the planning horizon.
- It is assumed that only AC appliances are used and are connected to the load bus.
- A time horizon of 1 h is used throughout this study.
- It is further assumed that the village electrical load and the renewable sources are constant within each one-hour time step.

The unit of measurement for power is kW and that for electrical energy is kWh. It is assumed that all the system components shown in Fig. 1 are installed at a specific site.

4.2. Objective function

The main objective function to determine the optimum cost of a hybrid energy system is expressed as:

$$\text{Minimize}: \quad \text{TC} = \sum_{d=1}^{d_n} \sum_{j=1}^{6} \sum_{t=1}^{24} \left[C_j \times E_{jdt} \right] \tag{12.1}$$

where, TC is the total optimized cost of providing energy for all end uses for operation of the system; C_j cost/unit of the jth generating unit (Rs./kWh); E_{jdt} optimal amount of the energy of the generating unit j for end use in a day d, hour t for a particular month; d_n is number of days depending upon a particular month.

In addition to that, the diesel engine generator is desired to be operated under constant output with high efficiency in order to reduce the polluting gases from the diesel engine; therefore, other secondary objectives of the model are as follows:

1. Keep the output of the diesel generator constant with high efficiency.
2. Minimize the fuel consumption of the diesel generator.
3. Minimize the frequency of diesel generator starts/stops.
4. Maximize the utilization rate of the renewable energy sources.

4.3. Integer variable constraints

Integer variables are used for representing the operation or otherwise of the different renewable generators, diesel generator, and battery bank. Let X_{MHG}, X_{BGG}, X_{BMG}, X_{PVG}, X_{DEG}, and X_{BATT} be the 0–1 integer variables representing the decisions to select or not select MHG, BGG, BMG, PVG, DEG and BATT components respectively, of the hybrid energy system. That is,

$$X_j = \left\{ \begin{array}{ll} 1 & \text{If unit } j \text{ serves the load directly} \\ 0 & \text{Otherwise} \end{array} \right\} \tag{12.2}$$

4.4. General constraints

The optimization is to be performed, enforcing a set of restrictions, which can be classified in the following groups:

1. Energy balance constraint: In order to follow the system hourly load demand, the hourly energy generation/discharge sum of all on-line units must equal the system load demand over the whole scheduling time range. That is

$$E_{j,j\neq \text{BATT}}(t) \pm E_{\text{BATT}}(t) \geq E_{\text{Load}}(t) + E_{\text{Dump}}(t) \tag{12.3}$$

where $+E_{\text{BATT}}(t)$, $-E_{\text{BATT}}(t)$ is the energy supplied and used by the battery respectively.

2. Individual capacity constraint: The power supplied by each generator/discharge unit must be less than or equal to the capacity of the unit.

$$P_j(t) = \text{Cap}_j \quad \text{for all } t \tag{12.4}$$

3. Unit generation limits: Each generation/discharge unit has its maximum and minimum generation limit. Therefore, in order to avoid damages, the hourly generation/discharge of each unit must meet constraints as follows:

$$P_{j\,\min} \leq P_j(t) \leq P_{j\,\max} \quad \text{for all } t \tag{12.5}$$

4. Battery storage limits: The energy stored in batteries at any hour t is subject to the following constraints:

$$E_{\text{BATT min}} \leq E_{\text{BATT}}(t) \leq E_{\text{BATT max}} \quad \text{for all } t \tag{12.6}$$

$$E_{\text{BATT min}} = (1 - DOD) E_{\text{BATT max}} \tag{12.7}$$

This means that the batteries should not be over discharged or over charged at any time. That protects batteries from being damaged.

5. Renewable energy resource constraints: Renewable energy should be used as much as possible.
6. Non-negativity constraints: Since electrical energy flow and power flow can not be negative in the solution. Hence, all the decision variables are non-negative.

4.5. Decision variables and constraints associated with PVG

Decision variables of photovoltaic generator are $X_{j=\mathrm{PVG}}$, and $E_{jdt,j=\mathrm{PVG}}$ The former is an integer decision variable representing a decision to select or not select a photovoltaic generator (PVG) in an hour t; whereas the latter is a continuous decision variable representing power generation from photovoltaic generator in day d, and hour t of a particular month. The following equations represent the generation characteristics of PVG. They imply that the power generation from the PVG at any hour t can take the value at its maximum generation capacity, if the PVG is selected.

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) < [G(t) \times A \times P \times \eta_{\mathrm{PVG}}] \times X_{j=\mathrm{PVG}} \quad (12.8)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) + E_{jdt,\mathrm{BATT}}(t) + E_{jdt,\mathrm{Dump}}(t) = G(t) \times A \times P \times \eta_{\mathrm{PVG}}$$

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) \geq [G(t) \times A \times P \times \eta_{\mathrm{PVG}}] \times X_{j=\mathrm{PVG}} \quad (12.9)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) = G(t) \times A \times P \times \eta_{\mathrm{PVG}}$$

4.6. Decision variables and constraints associated with MHG

Similarly, the following equations represent the generation characteristics and constraints associated with MHG.

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) < \left[P_{j=\mathrm{MHG}}(t) \times e \times \eta_{\mathrm{MHG}}\right] \times X_{j=\mathrm{MHG}} \quad (12.10)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) + E_{jdt,\mathrm{BATT}}(t) + E_{jdt,\mathrm{Dump}}(t) = P_{j=\mathrm{MHG}}(t) \times e \times \eta_{\mathrm{MHG}}$$

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) \geq \left[P_{j=\mathrm{MHG}}(t) \times e \times \eta_{\mathrm{MHG}}\right] \times X_{j=\mathrm{MHG}} \quad (12.11)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) = P_{j=\mathrm{MHG}}(t) \times e \times \eta_{\mathrm{MHG}}$$

4.7. Decision variables and constraints associated with BGG

The generation characteristics and constraints associated with BGG are given below.

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) < \left[P_{j=\mathrm{BGG}}(t) \times \eta_{\mathrm{DFEG}}\right] \times X_{j=\mathrm{BGG}} \quad (12.12)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) + E_{jdt,\mathrm{BATT}}(t) + E_{jdt,\mathrm{Dump}}(t) = P_{j=\mathrm{BGG}}(t) \times \eta_{\mathrm{DFEG}}$$

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) \geq \left[P_{j=\mathrm{BGG}}(t) \times \eta_{\mathrm{DFEG}}\right] \times X_{j=\mathrm{BGG}} \quad (12.13)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) = P_{j=\mathrm{BGG}}(t) \times \eta_{\mathrm{DFEG}}$$

4.8. Decision variables and constraints associated with BMG

The generation characteristics and constraints associated with BMG are expressed as:

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) < \left[P_{j=\mathrm{BMG}}(t) \times \eta_{\mathrm{DFEG}}\right] \times X_{j=\mathrm{BMG}} \quad (12.14)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) + E_{jdt,\mathrm{BATT}}(t) + E_{jdt,\mathrm{Dump}}(t) = P_{j=\mathrm{BMG}}(t) \times \eta_{\mathrm{DFEG}}$$

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) \geq \left[P_{j=\mathrm{BMG}}(t) \times \eta_{\mathrm{DFEG}}\right] \times X_{j=\mathrm{BMG}} \quad (12.15)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) = P_{j=\mathrm{BMG}}(t) \times \eta_{\mathrm{DFEG}}$$

4.9. Decision variables and constraints associated with DEG

The following equations represent the generation characteristics and constraints associated with DEG. They imply that the power generation from the DEG at any hour t can take the value zero, or any value between its minimum generation (which is assumed to be 80% of its rated power), and its maximum generation capacity (which is its rated power), if the DEG is selected.

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) < \left[P_{j=\mathrm{DEG}}(t) \times \eta_{\mathrm{DEG}}\right] \times X_{j=\mathrm{DEG}} \quad (12.16)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) + E_{jdt,\mathrm{BATT}}(t) + E_{jdt,\mathrm{Dump}}(t) = \left[0.8P_{j=\mathrm{DEG}}(t), 1.0P_{j=\mathrm{DEG}}(t)\right] \times \eta_{\mathrm{DEG}}$$

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) \geq \left[P_{j=\mathrm{DEG}}(t) \times \eta_{\mathrm{DEG}}\right] \times X_{j=\mathrm{DEG}} \quad (12.17)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) = \left[1.0P_{j=\mathrm{DEG}}(t)\right] \times \eta_{\mathrm{DEG}}$$

This research also assumes that diesel generator must be turn off after t = 24 to minimize noise in the early hours of the day by meeting the load using battery as much as possible. Finally, diesel generator should have time limits of operation to reduce wear and tear.

$$\sum_{t=1}^{24} X_{j=\mathrm{DEG}} \leq T_{\mathrm{DEG}} \quad (12.18)$$

$$0 \leq T_{\mathrm{DEG}} \leq K \quad (12.19)$$

4.10. Decision variables and constraints associated with BATT

Similarly, the generation characteristics of BATT implies that the discharging from the battery at any hour t can take the value zero, or any value between its minimum discharge capacity (which is assumed to be 20% of its rated capacity), and its maximum discharge capacity (which is its battery rated capacity), if the battery is selected.

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) < \left[P_{j=\mathrm{BATT}}(t) \times \eta_{\mathrm{DCHG}}\right] \times X_{j=\mathrm{BATT}} \quad (12.20)$$

$$\text{i.e.} \quad E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t)/(\eta_{\mathrm{INV}} \times \eta_{\mathrm{DCHG}}) = \left[0.2P_{j=\mathrm{BATT}}(t), 1.0P_{j=\mathrm{BATT}}(t)\right] \times \eta_{\mathrm{DCHG}}$$

$$E_{jdt}(t) = E_{jdt,\mathrm{Load}}(t) \geq \left[P_{j=\mathrm{BATT}}(t) \times \eta_{\mathrm{DCHG}}\right] \times X_{j=\mathrm{BATT}} \quad (12.21)$$

$$\text{i.e.} \quad E_{jdt}(t) = 0$$

Lastly, in order for the system with battery to be sustained over a long period of time, the battery SOC at the end of the day must be greater than a percentage of its $\mathrm{SOC}_{\max}$. So that it can be operated in

the early hours of the next day. This study assumes 80% as shown below:

$$SOC(t = 24) \geq 0.8\ SOC_{max} \tag{12.22}$$

5. Proposed combined dispatch strategy

The dispatch strategy for a hybrid energy system is a set of rules for the interaction among various system components. It determines the energy flows from the various sources to the load including the charging and discharging of the energy storage systems, on a time scale of minutes to hours, in such a way as to optimize system performance in terms of operating cost.

In a system that has different renewable energy sources, the output from these renewable generators is generally subtracted from the hourly total load demand to determine the hourly netload. By observing hourly operation of proposed hybrid energy system configuration, there are five possible dispatch strategies to meet the netload [3,10]. If netload is zero or negative in a particular hour then battery charging strategy is used to absorb the surplus power (all or a fraction), generated by renewable generators. Alternatively, if netload is positive, then all five dispatch strategies are used to operate and control the system. The summarized strategies that are modelled in this study are described below:

1. Battery charging strategy: The use of only the battery to absorb the surplus power. The absorption of energy continues until:
   - Maximum battery SOC (or 80% of rated capacity) is reached.
   - The renewable power is not sufficient to meet the load.
2. Battery discharging strategy: Battery energy may be used to meet the netload in a time step. The battery discharging continues until:
   - Minimum battery SOC for discharge is reached.
   - The renewable power is sufficient to meet the load.
   - The renewable power is sufficient to meet the load as well as continue to charge the batteries.
   - Netload (or netload + battery load) is equal to or greater than diesel minimum operating power.
3. Load following strategy: The diesel generator is ON to follow the netload, no charge to the battery and no discharge from the battery. A minimum diesel run time is also be applied to avoid excessive start/stop frequency.
4. Cycle charging strategy: The diesel generator is ON to cover the netload demand and charge the battery. The diesel continues running for its prescribed minimum run time; after that, the diesel continues running until one of the condition is met:
   - The prescribed SOC set point has been met, or
   - The renewable power is sufficient to meet the load.
   - The renewable power is sufficient to meet the load as well as continue to charge the batteries.
5. Peak shaving strategy: Diesel will operate at full power. Battery power is only used to meet buffer instantaneous fluctuations around the netload.

Let the total energy generated by renewable generators in the system at the time t, $E_{REG\text{-}Total}(t)$, can be expressed as follows:

$$E_{REG\text{-}Total}(t) = E_{MHG}(t) + E_{BGG}(t) + E_{BMG}(t) + E_{PVG}(t) \tag{13.1}$$

and the hourly netload demand ($E_{Netload}$) remaining for diesel generators and battery bank is given by

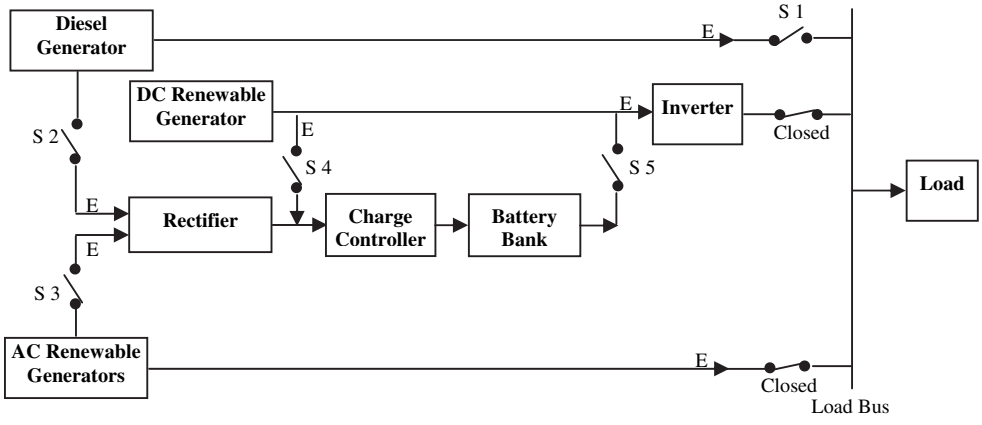
$$E_{Netload}(t) = E_{Load}(t) - E_{REG\text{-}Total}(t) \tag{13.2}$$

This system has three conditions of operation to supply the netload, which are given below:

5.1. Condition-1[$E_{REG\text{-}Total}(t) \geq E_{Load}(t)$]

When the energy generated from renewable generators (MHG, BGG, BMG, and PVG) exceeds the load demand for the system, the excess energy will charge the battery bank (provided they are not already fully charged) as shown in Fig. 2. Opportunities for this type of charging occur frequently in system with high solar insolation (battery charging strategy).

There are two possibilities in this condition. In the first case, the battery has not reached the maximum capacity and the excess power will charge the battery bank. In other case, the excess power



Battery charging strategy	Switches S3, S4 Closed, S1, S2, S5 Open
Battery discharge strategy	Switches S5 Closed, S1, S2, S3, S4 Open
Load following strategy	Switches S1 Closed, S2, S3, S4, S5 Open
Cycle charging strategy	Switches S1, S2 Closed, S3, S4, S5 Open
Peak shaving strategy	Switches S1, S5 Closed, S2, S3, S4 Open

Fig. 2. Diagram of energy flow of proposed strategy.

(all or some fraction) will be lost because the battery bank is fully charges.

5.2. Condition-2[$E_{REG\text{-}Total}(t) < E_{Load}(t)$] & [$E_{Netload}(t) < E_{DEG}(t)$]

When the total energy generated from renewable generators are lower than load demands, then first battery bank will be discharged (because unit cost of battery bank is lower than the unit cost of diesel generator) by the amount necessary to cover the netload and the diesel generator turns off (battery discharge strategy) as shown in Fig. 2.

If the batteries are not able to supply such a netload completely, or the strategy does not allow it, then the netload is met by diesel generator. There are two possible dispatch strategies in this condition. These are the cycle charging strategy and the load following strategy. The cycle charging strategy, which is discussed above, is generally used in combination with the load following strategy. The decision to use one or the other strategy, i.e., the decision to generate power equal to the netload (load following) or greater than the netload to charge the battery (cycle charging), can be arbitrary selected as shown in Fig. 2.

When the netload on the diesel generator is low, the diesel output can be increased, with the excess power produced being used to charge the battery storage. The advantages of cycle charging are that it allows the diesel generator to operate at higher power output, where it operates more efficiently; and it allows for longer diesel shutdown periods, thus reducing the rotating cost associated with diesel generators.

5.3. Condition-3[$E_{REG\text{-}Total}(t) < E_{Load}(t)$] & [$E_{Netload}(t) > E_{DEG}(t)$]

If the netload exceeds the diesel generator rated capacity, the diesel generator will run at full power and batteries will attempt to contribute the difference as shown in Fig. 2. This type of simultaneous operation of both battery and diesel generator is known as peak shaving strategy.

6. System control

The control system manages the energy flow among the different components of the hybrid energy system. Here, it is assumed that the control system have the means of estimating the battery SOC and netload at each hour. Battery SOC and netload are used as parameter for the ON/OFF control of diesel generator and the disconnection/reconnection of the load. The system energy balance (i.e. netload) and battery SOC are checked at each time step of the simulation. Checking the energy balance and battery SOC at each time step assures that the model and simulation algorithm are internally consistent.

Following requirements must be met by the control system:

1. Direct energy flow from different generators to the load should be maximize in order to avoid losses in the buffer energy storage i.e. batteries.
2. All types of renewable generators, diesel generator and battery bank may operate simultaneously, if needed.
3. Once started the diesel engine generator should run for a minimum time period, typically exceeding at least 20 min of continuous operation. This performance specification is implemented to reduce engine wear and to minimize maintenance requirements.
4. The diesel generator should not be allowed to operate below a minimum power level to optimize the fuel efficiency.

Thus, some predefined conditions are programmed in the developed algorithm.

7. Implemented economic dispatch algorithm

The economic dispatch problem concerns the question of how to distribute a load demand over the units that are in service so that the total cost of operation is at a minimum.

In common solution of Economic dispatch as can be seen in the literature review, when the load is light, the cheapest generating units are always the ones chosen to run first. As the load increases, more and more expensive generators will be brought in. Thus, the operating cost (or unit cost) plays a very important role in the solution of Economic Dispatch.

The energy output models for various generating/discharge units developed in the previous sections are used for developing the economic dispatch algorithm. An algorithm has been written in C++ language, in order to ensure modularity and easily modify the system models, in terms either of parameters or control strategy.

The simulation algorithm (or program) can be broadly divided into three sections: The database (or Input), supporting database consists of different constraints and control set points and output system data file. The developed algorithm provides sufficient information about the performance of each component and the system. During the program execution, all necessary system variables needed for a performance analysis are recorded in a simulation output file. For example, total optimum cost, optimized system unit cost, diesel run hours/day, number of diesel starts/stops, unmet energy (if any), dump energy etc.

Objective function and constraints have been already defined in section 4. The other details of the simulation algorithm are given below:

7.1. Assumptions

1. It is assumed that the shape of load curve does not change throughout the planning period.
2. Each renewable generator is assumed to have two possible states: (a) zero outage level (generator is running at full capacity), (b) full outage level (generator is out).
3. (a) Loading of renewable generators is done according to economic merit order, that is, generators are loaded in order of increasing unit generation cost. (b) If the renewable generators are not able to supply such an hourly total load demand, then battery bank and diesel generator will be loaded according to economic merit order.
4. It is assumed that the same economic order is preserved throughout the planning period.

7.2. Database

The essential data for running the simulation algorithm are collected, organized, and stored in a database. The information stored in the database includes:

- Available renewable energy generator power production capacity.
- Hourly solar insolation.
- Hourly demand load profile of the cluster of villages.
- Diesel generator and battery bank capacity.
- Hourly scheduling of different renewable energy generators.
- Unit cost of generation of different energy sources.
- Minimum and maximum SOC of battery bank.

7.3. Control parameters

Several key system control parameters also needed to be supplied to the program which is given in Table 1.

7.4. Optimal economic operation strategy

A computer program is developed to determine the optimum unit cost of hybrid energy system based on energy balance and combined dispatch strategy during a day and summed to give the overall performance over a month. Each day is divided into a 24 one hour segments, with the energy generated and consumed during each segment, taking into account both economic and technical factors.

Figs. 3–7 describe the computer algorithm to simulate the operation of the desired hybrid energy system. The following items summarize the key characteristics of the implemented strategy. According to Fig. 3:

1. Renewable power (MHG, BGG, BMG, and PVG) is first sent to meet the village load demands in order of economic merit. Energy output of each REG is calculated according to equations developed in Section 3. If the power supply from REGs is smaller than the load demand, then supply the load equal to maximum operating power from REGs and go to step 3. Alternatively, if the power supply from REGs is larger than the power demand, then supply the load and go to step 2.
2. Charge all surplus power in battery. Here, if battery becomes full, charge surplus power until battery become full and remaining surplus power is thrown out (goes to dump load). Go to step 5.
3. Energy output from renewable generators is treated as a negative load. So the netload can be determined according to Eq. (13.2). If the netload is equal to zero, then go to step 5. If the netload is positive, then go to step 4.
4. If the system has battery storage and diesel generator both, the operation characteristic will be divided according to time that the diesel generator starts or stops. The following description explains the operation of the diesel generator during $t = 0$ to $t = T_{start}$; during $t = T_{start}$ to $t = T_{stop}$; and during $t = T_{stop}$ to $t = 24$ of each day.
   - 4.1 During $t = 0$ to $t = T_{start}$ of each day: if netload is less than diesel generator minimum operating power, then the system is operated only on the battery.
     - 4.1.1 Energy discharged from the battery must be enough to serve hourly netload demand, i.e. netload/($\eta_{DCHG} \times \eta_{inv}$), but not be greater than the maximum allowable discharged energy from the battery.
     - 4.1.2 Unmet energy is zero if the discharge energy from the battery can cover the netload.
     - 4.1.3 If battery is not sufficient to meet the netload and diesel generator maximum running hours has completed, then there will be a portion of demand unmet.
     - 4.1.4 SOC of the battery at any hour t is determined according to eq.10.2 and 10.3.
     - 4.1.5 No energy is generated from diesel generator during this period. However, whenever SOC reaches SOC_{min}, the diesel generator must be started in order to cover the netload demand and charge the battery (i.e. go to step 4.2), otherwise go to step 5.
   - 4.2 During $t = T_{start}$ to $t = T_{stop}$ of each day: This section is divided into four parts -
     - 4.2.1 If diesel generated power is greater than netload and netload is greater than diesel generator minimum operating power, then the system is operated on the diesel generator to cover the netload and to charge the battery. Go to step 5.
     - 4.2.2 If diesel generated power is greater than netload and netload is smaller than diesel generator minimum operating power (Vt), then check:
       Netload + battery load $\geq$ Vt
       If yes, than go to step 4.2.1, otherwise check
       $t = 24$ and SOC $< 80\%$
       If yes, than go to step 4.2.1, otherwise go to step 4.1.
     - 4.2.3 If diesel generated power is smaller then netload, then the system is operated on diesel generator and battery both. Diesel generator operates at maximum operating power to cover the maximum part of netload and rest part is covered by battery. Go to step 5.
     - 4.2.4 If diesel generator minimum running hours has completed or SOC of battery has reached up to 80% or more, then stop diesel generator and go to step 4.3.
   - 4.3 During $t = T_{stop}$ to $t = 24$ of each day: the system is operated from the battery, so the algorithm is the same as that during $t = 0$ to $t = T_{start}$ of each day.
5. If time t does not reach the end of simulation, increase $t = t + 1$ and go to step 1. If time t reaches end of simulation, go to End.

Table 1
Control parameters for simulation.

S.N.	Parameters	Set value
1.	Minimum battery SOC control point (SOC_{min})	0.20
2.	Maximum battery SOC control point (SOC_{max})	1.00
3.	Diesel generator minimum loading	0.80
4.	Diesel generator maximum loading	1.00
5.	Diesel generator starting point (fraction of battery capacity)	0.20
6.	Diesel generator starting point (fraction of netload)	36.8 kWh
7.	Diesel generator stopping point (fraction of battery capacity)	0.80
8.	Diesel generator stopping point (fraction of netload)	0.0 kWh

8. Case study

This study aims at providing a detailed techno-economic evaluation of proposed hybrid energy system to supply electricity to selected rural remote villages.

8.1. Study area

Depending on a comprehensive assessment on non-electrified villages in India [11], Narendra Nagar block of district Tehri Garhwal, Uttarakhand State has been selected as remote area. The block consists of nine un-electrified villages with 13 hamlets. Villages are situated between latitude 30°07′ N and 30°21′ N and longitude 78°18′ E and 78°29′ E. There are 502 households with a population of 3223 [11]. The area comprises of major hilly and the fertile area under forest with scattered households. The area has been considered by Uttarakhand Renewable Energy Development Agency (UREDA) to be remote and not economically viable for electrification by grid extension.

8.2. Assessment of energy potential

The study area occupies a unique position as far as natural sources are concerned.

i. The solar radiation in this area is on the average 5.08 kWh/m^2 and goes a high as 6.06–6.95 kWh/m^2 in the months from

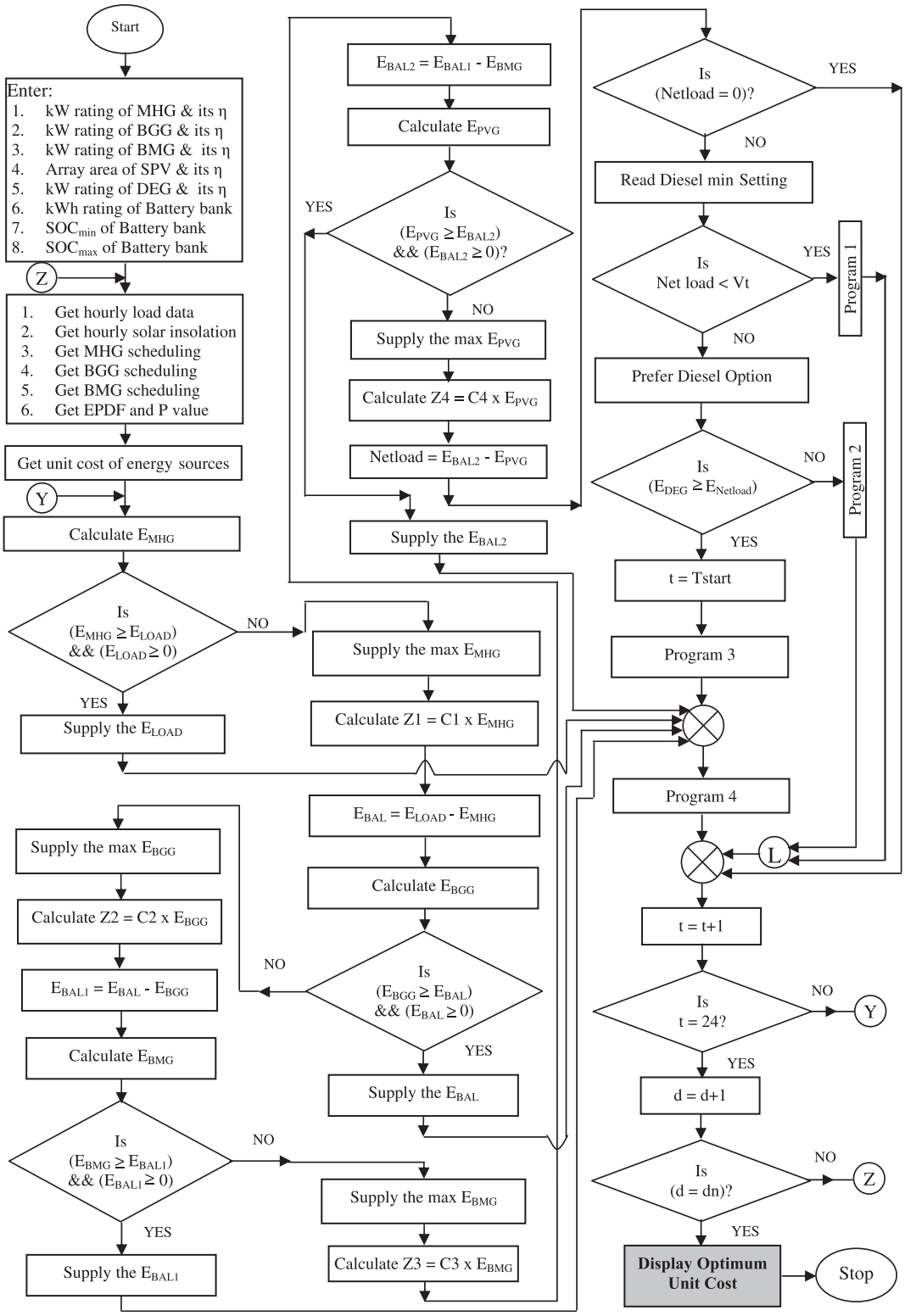


Fig. 3. Flow chart for main program.

April to June. Annual total solar energy potential available is 1854.18 kWh/m^2/yr [12].

ii. It has found that out of nine villages, only three villages have micro hydro potential. The total potential at these sites has been estimated as 14.2 kW (~ 15 kW) [11].

iii. To assess the biogas potential, buffaloes, horse, goats, and cow/ox have been considered. Based upon the survey, it was found that there is 4564 cattle population at study area. The biogas production from the dung has been evaluated based on the assumption that 10 kg/day dung will be available from cow/ox, 15 kg/day from buffaloes, 1 kg/day from goat, and 10 kg from horse. The cattle dung availability in the study area is about 25,560 kg/day. The biogas estimation is based on the cattle dung production from different types of animals and

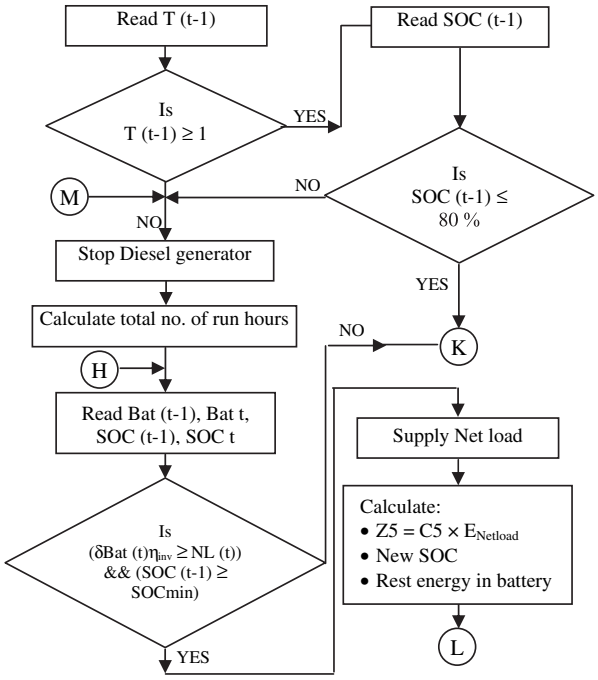


Fig. 4. Flow chart for program 1.

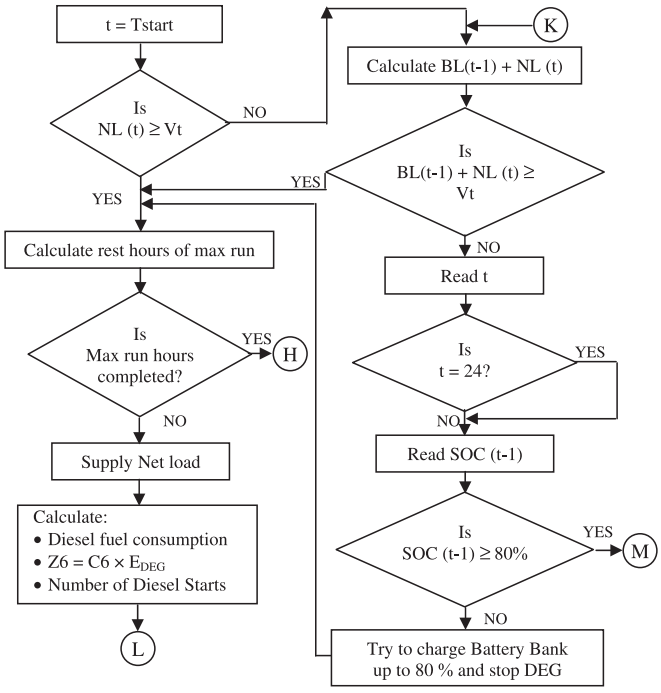


Fig. 6. Flow chart for program 3.

assuming that 0.036 m^3 of biogas is generated per kg of cattle dung. Therefore, the biogas availability in the study area is about 644.112 m^3/day out of which biogas 510.8796 m^3/day, is used for cooking. The balance 133.2323 m^3/day is available for generation of electricity.

iv. To assess the biomass potential, agricultural and forest waste (fuelwood) have been considered. It is estimated that about 1083.35 Ton/yr fuelwood and 44.71 Ton/yr of crop residue is available as surplus by taking 2% sustainable yield of fuelwood [11]. However, in this study only 1% sustainable yield of fuelwood is considered to avoid deforestation and crop residues are left to feed the livestock. The balance 147.757 Ton/yr is available for generation of electricity.

The village wise availability of micro-hydro, biogas and biomass is shown in Table 2.

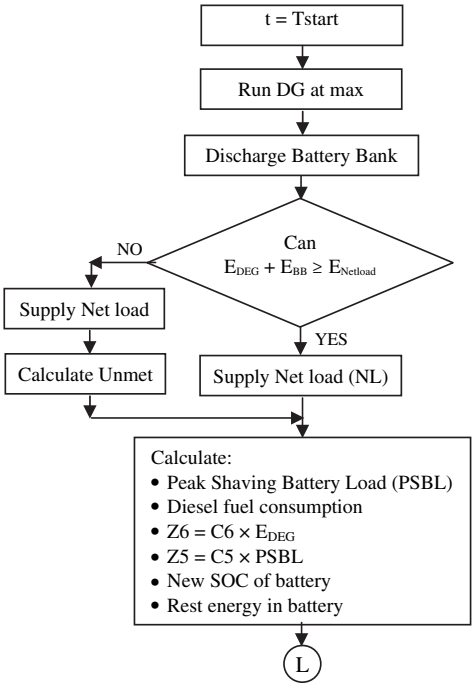


Fig. 5. Flow chart for program 2.

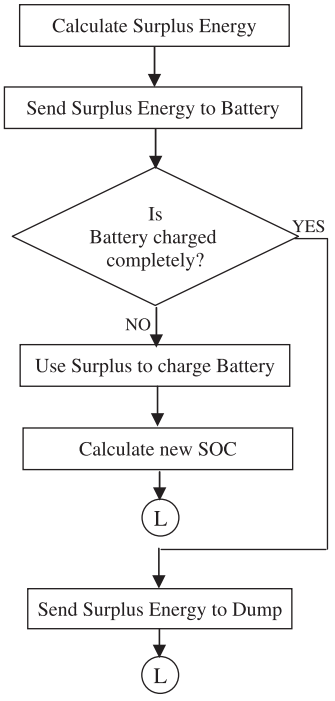


Fig. 7. Flow chart for program 4.

Table 2
Details of energy resource potential.

S.N.	Village name	Micro-hydro availability[a] (h/Q)	Biogas availability (m^3/day)	Biomass availability (Ton/yr)
1.	Laga Mehra	–	77.112	–
2.	Saud	–	83.412	–
3.	Salem Khet	–	21.42	–
4.	Talai lambadi	2 (7.00/0.05)	38.556	–
5.	Bandhan	–	15.372	–
6.	Pungarh	7.00/0.06	13.86	–
7.	Bhangla	–	70.812	98.99
8.	Kakhoor	–	79.632	48.76
9.	Banskata	5.00/0.06	243.936	–
Total potential		–	**644.112**	**147.757**

[a] Available only in Mar–Oct period by consulting senior citizens.

Table 4
Data used for simulation.

S.N.	Type of energy resources	Cost of energy (Rs./kWh)	Efficiency	Daily operating periods
1.	Micro-hydro generator	1.45	0.60	22 h
2.	Photovoltaic generator	15.68	0.1154	–
3.	Biogas generator	3.98	1.0	10 h
4.	Biomass generator	4.78	1.0	12 h
5.	Diesel generator	11.0	1.0	10 h (max)
6.	Battery	3.26	0.90,[a] 1.0[b]	–
7.	Inverter/rectifier	–	0.95	–
8.	PVG-inverter	17.72	–	–
9.	Battery-inverter	4.33	–	–
10.	Charge controller	–	0.90	–

[a] Charging efficiency
[b] Discharging efficiency

8.3. Demand assessment

The data for load demand estimation has been collected on the basis of questionnaire. The energy demand is estimated by considering the household load, commercial load, industrial load and community load. Demand side management (DSM) is also obtained by using 11 W, and 20 W compact fluorescent lamps (CFL), and 55 W energy saver fans instead of 60 W, 100 W incandescent lamps and 65 W ordinary fans respectively. After the replacement, the total daily energy demand of cluster of nine village's drops to 1271.61 (32.30% decrease) and 608.97 kWh/day (44.40% decrease) in summer and winter respectively. Similarly peak load is reduced to 113.824 (37.65% decrease) and 68.774 kWh/day (41.70% decrease) in summer and winter respectively. By summing up all appliance demands in each time interval, the overall load profile can be made. Table 3 shows the description of the different categories of load, operating period of different electric appliances, and the corresponding electrical energies.

8.4. Results and analysis

Based on the time series calculations, the developed model simulates the hybrid energy system, so that, for estimated load

Table 3
Details of total load of nine villages.

Time segment (hours)	Electrical load (kWh)										Total electrical load/hour (kWh) summer/winter
	Household load				Commercial load		Industrial load	Community load			
	Lighting Load (11 W)	T.V. (90 W)	Fan (55 W) summer/ winter	Radio/music system (25 W)	Lights for shops (20 W)	Floor mill (5 kW)	Saw mill/paddy huller (5 kW)	Primary health centre (20 W)	Street lights (20 W)	School lights (20 W)	
0:0–1:0									2.0		2.0/2.0
1:0–2:0									2.0		2.0/2.0
2:0–3:0									2.0		2.0/2.0
3:0–4:0									2.0		2.0/2.0
4:0–5:0	5.522								2.0		7.522/7.522
5:0–6:0	5.522								2.0		7.522/7.522
6:0–7:0	11.044										11.044/11.044
7:0–8:0	11.044	45.18		12.55							68.774/68.774
8:0–9:0		45.18		12.55				0.040		0.72	58.49/58.49
9:0–10:0						5.0	5.0	0.040		0.72	10.76/10.76
10:0–11:0						5.0	5.0	0.040		0.72	10.76/10.76
11:0–12:0			55.22/0.0			5.0	5.0	0.040		0.72	65.98/10.76
12:0–13:0		45.18	55.22/0.0			5.0	5.0	0.040		0.72	111.16/55.94
13:0–14:0		45.18	55.22/0.0								100.40/45.18
14:0–15:0			55.22/0.0								55.22/0.0
15:0–16:0			55.22/0.0	12.55							67.77/12.55
16:0–17:0			55.22/0.0	12.55							67.77/12.55
17:0–18:0	11.044	45.18	55.22/0.0		0.34			0.040			111.824/56.604
18:0–19:0	11.044	45.18	55.22/0.0		0.34			0.040	2.0		113.824/58.604
19:0–20:0	11.044	45.18	55.22/0.0		0.34			0.040	2.0		113.824/58.604
20:0–21:0	11.044	45.18	55.22/0.0		0.34			0.040	2.0		113.824/58.604
21:0–22:0	5.522	45.18	55.22/0.0						2.0		107.922/52.702
22:0–23:0			55.22/0.0						2.0		57.22/2.0
23:0–24:0									2.0		2.0/2.0
Daily load	**82.83**	**406.62**	**662.64/0**	**50.20**	**1.36**	**20.0**	**20.0**	**0.360**	**24.0**	**3.60**	**1271.61/608.97**
Electrical appliances used[a]	**1–2/ 1004**	**1/502**	**2/1004**	**1/502**	**1/17**	**1/1**	**1/1**	**1/2**	**1/100[b]**	**1/36**	–
Contribution (%)	**93.42**				**2.03**		**1.90**	**2.65**			**100**

[a] (/) Quantity per HH or room or in area/total
[b] 1 pole @ clusters of 5 households.

Table 5
Optimized configurations as a function of the solar photovoltaic contribution.

S.N.	Penetration level PV (%)	PV Area (m^2)	Battery size (kWh)	DEG unit (kW)	Unit cost (Rs./kWh)	Dump energy (%)	Unmet energy (%)
1.	0	0	0	51	10.41	0	0
2.	**20**	**201.8507**	**106**	**46**	**10.08**	**0.35**	**0**
3.	40	403.7014	148	35	10.85	1.64	0.12
4.	60	605.5521	194	23	11.27	1.27	0
5.	80	807.4028	276	12	10.78	0.41	0
6.	100	1009.2356	373	0	9.7619 (16.6993)	0	0

Table 6
System component sizing results.

S.N.	Type of components	Installed capacity (kW)
1.	MHG	15
2.	PVG	202 m^2
3.	BGG	20
4.	BMG	34
5.	DEG	46
6.	Battery	106 kWh
7.	Inverter	35
8.	Dump load	12

profile, the fuel consumption along with energy share and other economical & operational data are calculated.

One set of 24 h data (site load, solar potential and scheduling of renewable generators) are gathered in each month from January to December. These 12 sets of data are used in simulation algorithm as an input data to simulate the output. Different values for parameters used for simulation purpose is shown in Table 4.

8.4.1. Simulation results for the assessment of the economic penetration level of PV array area and optimal configuration

The validity of the model and proposed strategy is examined by applying it to a hybrid generation system having the data given in Table 4. Firstly, the generating capacity for MHG, BGG, and BMG are determined on the basis of respective potential assessment using equations given in appendix. Then the required generating capacity of PV array area is determined according to the amount of rest total daily electrical energy demand for design month. After this, the generating system is designed in different proportion (penetration) of PV array area: only 0% proportion, only 20% proportion, 40% proportion, 60% proportion, 80% proportion, and 100% proportion. Hourly solar radiation data for this site have been calculated at latitude 30°32′ N, Longitude 78°03′ E [12].

The model and operation strategy is applied to the hybrid generation system for the six cases given in Table 5. The simulation program will use above mentioned input and repeatedly simulates hourly system operation over the month. For each combination, the hourly, daily and monthly unit costs are evaluated. The feasible solutions are ranked by system optimized unit cost and are presented in Table 5, which clearly indicates that the economic optimal penetration level is 20% for photovoltaic array area. Also the optimum unit sizing of different components, for the design month August is presented in Table 6. The hourly detailed simulation operation as a sample for case 2 (i.e. 20% penetration level) is given in Table 7.

It is evident that there are different factors affecting the economic penetration level of PV, battery bank and diesel generator capacity. The main factors are solar radiation profile of the site and economic considerations imposed on the problem. A slight change in any of these factors has a significant effect on the results derived.

8.4.2. Simulation results for cost optimization

The results from the simulation for the optimal operation of the system during a typical day (i.e. 2nd day of design month August) are presented in tabular form in Table 7.

The second column of Table 7 gives the total hourly load demand for 24 h; it had a total load demand of 1271.61 kWh. The maximum hourly load demand is 113.824 kWh at 1800–2100 h and the minimum is 2 kWh at 2300–2400 and 0000–0400h.

The scheduling of micro-hydro generator, biogas generator, and biomass generator is given in third, fourth and fifth columns respectively.

Table 7
Hourly simulation results for design month August.

Time segment	E_{LOAD} (kWh)	E_{MHG} (kWh)	E_{BGG} (kWh)	E_{BMG} (kWh)	E_{PVG} (kWh)	$E_{PVG\text{-}INV}$ (kWh)	$E_{NETLOAD}$ (kWh)	$E_{SURPLUS}$ (kWh)	$E_{BATT\text{-}INV}$ (kWh)	$E_{BATT\text{-}LEFT}$ (kWh)	E_{DEG} (kWh)	DEG Status	E_{UNMET} (kWh)	E_{DUMP} (kWh)	Unit cost Rs../kWh
0:0–1:0	2.0	–	–	–	0.0	–	2.0	0.0	2.0	91.7130	–	–	–	–	4.33
1:0–2:0	2.0	2.0	–	–	0.0	–	0.0	6.65	0.0	97.0995	–	–	–	–	1.45
2:0–3:0	2.0	2.0	–	–	0.0	–	0.0	6.65	0.0	102.4860	–	–	–	–	1.45
3:0–4:0	2.0	2.0	–	–	0.0	–	0.0	6.65	0.0	106.0	–	–	–	2.0806	1.45
4:0–5:0	7.522	7.522	–	–	0.0	–	0.0	1.4041	0.0	106.0	–	–	–	1.2637	1.45
5:0–6:0	7.522	7.522	–	–	0.0	–	0.0	1.4041	0.0	106.0	–	–	–	1.2637	1.45
6:0–7:0	11.044	9.0	–	–	2.0979	1.9930	0.0509	0.0	0.0509	105.9464	–	–	–	–	4.40
7:0–8:0	68.774	9.0	20.0	–	5.3615	5.0934	34.6805	0.0	34.6805	69.4406	–	–	–	–	4.84
8:0–9:0	58.49	9.0	20.0	–	8.8581	8.4151	21.0748	0.0	21.0748	47.2566	–	–	–	–	5.69
9:0–10:0	10.76	9.0	–	–	11.4223	1.7600	0.0	7.7514	0.0	55.0080	–	–	–	–	4.11
10:0–11:0	10.76	9.0	–	–	13.2872	1.7600	0.0	9.2620	0.0	64.2700	–	–	–	–	4.11
11:0–12:0	65.98	9.0	20.0	34.0	14.9189	2.9800	0.0	9.5435	0.0	73.8135	–	–	–	–	4.67
12:0–13:0	111.16	9.0	20.0	34.0	14.9189	14.1730	33.9870	0.0	33.9870	38.0377	–	–	–	–	5.88
13:0–14:0	100.40	9.0	20.0	34.0	13.2872	12.6228	24.7771	20.1618	0.0	54.3687	24.7771	On-Run	–	–	7.48
14:0–15:0	55.22	9.0	–	34.0	11.4223	10.8512	1.3688	42.3996	0.0	88.7124	1.3688	Run-Off	–	–	6.93
15:0–16:0	67.77	9.0	–	34.0	9.0912	8.6367	16.1334	0.0	16.1334	71.7298	–	–	–	–	5.88
16:0–17:0	67.77	9.0	–	34.0	5.1284	4.8720	19.8980	0.0	19.8980	50.7846	–	–	–	–	5.14
17:0–18:0	111.824	9.0	20.0	34.0	2.0979	1.9930	46.8310	0.0	0.8310	49.9098	46.0	On-Run	–	–	7.16
18:0–19:0	113.824	9.0	20.0	34.0	0.0	–	50.824	0.0	4.824	44.8319	46.0	Run	–	–	6.87
19:0–20:0	113.824	9.0	20.0	34.0	0.0	–	50.824	0.0	4.824	39.7540	46.0	Run	–	–	6.87
20:0–21:0	113.824	9.0	20.0	34.0	0.0	–	50.824	0.0	4.824	34.6761	46.0	Run	–	–	6.87
21:0–22:0	107.922	9.0	20.0	34.0	0.0	–	44.922	1.0241	0.0	35.5056	44.922	Run	–	–	6.94
22:0–23:0	57.22	9.0	–	34.0	0.0	–	14.220	30.1910	0.0	59.9603	14.220	Run	–	–	5.80
23:0–24:0	2.0	–	–	–	0.0	–	2.0	41.80	0.0	93.8183	2.0	Run-Off	–	–	11.0

Table 8
Details of renewable energy generation.

S.N.	Type of Renewable energy generators	Annual total (kWh)	Contribution (%)
1.	MHG output	48,510	15.82
2.	BGG output	72,225	23.55
3.	BMG output	142,620	46.50
4.	PVG output	43,369.09	14.14
5.	Total REG output	306,724.09	100

The sixth column shows the photovoltaic generator power profile, it had a maximum energy of 14.9189 kWh at 1100 to 1300 h and a minimum of 2.0979 kWh at 0600–0700 and 1700–1800 h. Similarly a column seventh, shows photovoltaic generator power after inverter operation.

The eighth and ninth columns give the total hourly netload (i.e. hourly deficit) and surplus energy for 24 h respectively. Similarly, other columns have the same meaning as indicated by the heading of the particular column.

All modelling and simulation is based on an hourly energy balance where the hybrid energy system supplies power to meet load demand (and possibly charge the batteries). For simplicity, the generator capacities were given as constant values, but the loads were varying. The initial battery energy level and the minimum allowable state of charge were set at 100% and 20% of maximum allowable state of charge respectively.

At day 1, the battery was started with initial SOC 106 kWh (100%), it was reached the minimum SOC of 34.6761 at 2000–2100 h, and at 2400, it had a SOC of 93.8183 kWh.

At day 2, battery bank was discharged with SOC of 93.8183 kWh, supplying the netload demand during hour 0000–0100 (battery discharge strategy).

The micro-hydro generator supplied the load demand completely during hour 0100–0600. The battery bank was charged completely by the micro-hydro generator (battery charging strategy). During this period the excess available energy was diverted to the dump load, which in this study was assumed to be electric water heater.

The load demand increased after 0600, at the same time, the solar photovoltaic contribution started to raise, the micro-hydro generator, photo-voltaic generator and batteries were used to supply the load demand (battery discharge strategy).

The load demand further increased after 0700, at the same time, biogas generator came into system, the micro-hydro generator, biogas generator, photovoltaic generator and batteries were used to supply the load (battery discharge strategy) during hours 0700–0900.

The load demand decreased after 0900, biogas generator stopped, and the micro-hydro generator and photovoltaic generator took over till the next generator was brought back online, the excessive energy from the solar photovoltaic was used to charge the battery bank (battery charging strategy) during hours 0900–1100.

The load demand increased after 1100, at the same time biomass generator came into system, biogas generator was brought back, and all the renewable generators and battery bank were used to supply the load.

Approximately, 1300 h, the solar power started to decrease; batteries were unable to supply the netload completely (battery control set point 1). At this time, diesel generator was brought online to meet the netload demand. The diesel was at its rated power. The excessive power generated from the diesel generator was used to charge the battery (cycle charging strategy). The battery SOC was going up till the diesel generator stopped operating, the battery bank attained the SOC 80% (or more) at the end of 1500 h, the diesel generator stopped, as specified in the set point (diesel generator stop set point 7), and the battery inverter took over till the diesel generator was brought back online.

The diesel generator was brought back online at approximately hour 1700, when the netload demand exceeded 36.8 kWh (diesel generator starts set point 6). The diesel generator supplied the netload and battery-inverter was used to shave the peak netload demand (peak shaving strategy). The netload demand decreased after 2100, the diesel generator was still operating due to set point constraints (diesel generator stop set point 7). The netload was supplied by the diesel generator. The excessive power generated from the diesel generator was used to charge the battery. The battery SOC was going up till the diesel generator stopped operating. The battery bank attained the 80% or more SOC at the end of 2400 h, finally, the diesel generator stopped, as specified in the control set point 7.

After this, the simulation profile was same for the rest days of the month, i.e., the battery was started with initial SOC of 93.8183 kWh, it was reached the minimum SOC of 34.6761 kWh, and at the end of the day, it had a SOC of 93.8183 kWh. In this way, the program repeatedly simulates hourly system operation over the month.

Once hourly energy output values of a stand-alone hybrid energy system are estimated, algorithm aggregates the hourly values to compute daily and monthly values. Daily energy values are used to perform a resource-load matching analysis aimed at evaluating an energy system's ability to meet the end-user's daily energy needs on an economic basis, while monthly values are used to evaluate system's performance. On the basis of monthly values, the annual value is employed to calculate a hybrid energy system's average cost of energy production.

Tables 8 and 9 gives a summary of the renewable energy generation and contribution of the various energy resources used in order to cover the load demand respectively, whereas Tables 10–12 gives a summary of unmet and dump energy, diesel engine generator performance and battery performance results respectively, and Table 13 gives detailed economic results.

Table 9
Details of Load Distribution.

S.N.	Load distribution between units	Annual total (kWh)	Contribution (%)
1.	Load by MHG	42,640.76	11.0864
2.	Load by BGG	70,685.27	18.3779
3.	Load by BMG	136,298.23	35.4370
4.	Load by PVG	27,633.55	7.1846
5.	Load by REG	277,257.61	72.0860
6.	Load by DEG	66,852.14	17.3810
7.	Load by BATT	40,510.77	10.5326

Table 10
Details of Unmet & Dump energy.

S.N.	Parameters	Annual total (kWh)	Annual average (daily)	Contribution (%) of total load
1.	Unmet energy	0.0	0.0	0.0
2.	Dump energy	3182.1376	8.72	0.8273

Table 11
Diesel generator performance results.

S. N.	Parameters	Annual total	Annual average (daily)
1.	Diesel fuel consumption (L)	25,396.2573	69.58
2.	Diesel run hours (h)	2129	5.83
3.	Diesel start–stops	337	0.92

Table 12
Battery performance results.

S.N.	Parameters	Annual total (kWh)
1.	Battery charge energy	32,191.3033
2.	Battery discharge energy	40,510.8066

Table 13
Economic results.

S.N.	Month	Parameters			
		Minimum hourly unit cost of HES (Rs./kWh)	Maximum hourly unit cost of HES (Rs./kWh)	Optimum daily unit cost of HES (Rs./kWh)	Optimum monthly unit cost of HES (Rs./kWh)
1.	Jan	3.98	17.72	5.65	5.65
2.	Feb	3.98	17.72	5.77	5.77
3.	Mar	1.45	11.0	6.35	6.35
4.	Apr	1.45	7.64	6.44	6.44
5.	May	1.45	11.0	6.45	6.45
6.	Jun	1.45	9.31	6.43	6.43
7.	Jul	1.45	11.0	6.40	6.40
8.	Aug	1.45	11.0	6.24	6.24
9.	Sep	1.45	11.0	6.28	6.28
10.	Oct	1.45	11.0	6.22	6.22
11.	Nov	3.98	17.72	5.75	5.75
12.	Dec	3.98	17.72	5.65	5.65
Annual Total Optimum Cost (Rs.)					**2,395,115.327**
Annual Total Load (kWh)					**384,620.85**
Annual Average daily Cost (Rs./kWh)					**6.23**

Rs. is the Indian currency, i.e. Indian rupees (INR).

The optimized annual system unit cost is determined by taking average of annual optimized total cost of all months, which comes out to be Rs. 6.23/kWh. The proposed HES can supply the entire cluster of village's population with 24 h quality power. Of the total primary energy requirement of the villages, the renewable generators produced 72.09% while the diesel generator and battery produced 17.38% and 10.53% of the energy.

The results shown are for specified parameters, which can vary for individual customers, as well as, from area to area. It can be seen that the least unit cost system is the stand-alone micro hydro generation system (1.45 Rs./kWh) as it has to be run almost all the time in order to meet the load demand constantly. On the other hand, the most expensive system is the stand-alone photovoltaic–inverter system (17.72 Rs./kWh). So, the stand-alone photovoltaic system will cost more money than it is necessary.

Regarding the biogas and biomass (fuelwood) based generation system it is clear that potential of biogas and biomass are sufficient with second (3.98 Rs./kWh) and third (4.78 Rs./kWh) lowest cost of energy. In order to fully utilize the biomass (biogas + fuelwood) resource, one is required to explore the possibility of generating electricity using dual fuel engine system.

Fig. 8 shows the optimization results of hybrid energy system for different values of EPDF (e) from 1.0 to 0.0. It gives the share of different sources with different EPDF value. It indicates that an EPDF of 1.0, the MHG plant deliver its maximum energy to the load.

Similarly, an EPDF of 0.9 gives 10% reduction in the energy delivery capability of the MHG plant. In the present model, it is felt also that the breakdown/non-functioning/fluctuation in the discharge of MHG plant may drastically affect the unit cost of hybrid energy system model. With decrease in EPDF from 1.0 to 0.0, the unit cost of generation increases from Rs. 6.23 to Rs. 7.17.

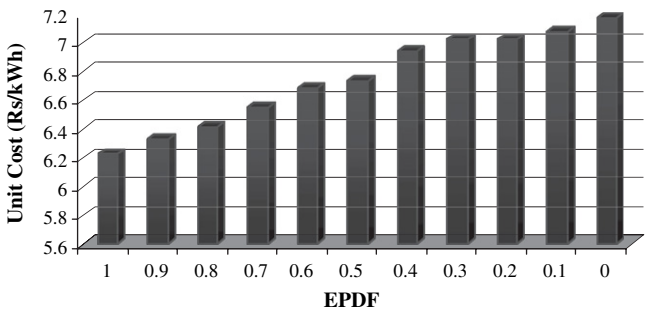


Fig. 8. Variation of unit cost with EPDF.

9. Conclusions

Taking into consideration the scientific interest concerning the capabilities of hybrid energy systems with energy storage to fulfill the electrification needs of a rural remote area, a detailed mathematical model and solution algorithm of describing the operational behavior of the basic hybrid energy system components is presented.

The algorithm presented is capable of efficiently designing a least-cost village electrification system while the diesel generator keeps the output constant with high efficiency in spite of the fluctuating photovoltaic power. Developed algorithm is modular in nature. The time-series optimized simulation is based on detailed component modelling. It can be extended to include other renewable sources of energy such as wind power and fuel cell.

Finally, a quite extensive case study is carried out, in order to demonstrate the validity of the model and proposed strategy. By using DSM, the hybrid energy system components are reduced in size and the energy available is utilized in a proper way, which can lead to reduction in the capital cost of the system. Further, PV energy is also utilized effectively (in different proportion of a PV array area) to optimize the size of the diesel generator and battery storage reducing the total system cost.

Taking into account the extensive results obtained, hybrid energy system may be cost-effective electrification solution for numerous isolated consumers. Our techno-economic feasibility study of hybrid energy system demonstrates that these systems can theoretically reduce generation costs and increase the reliability of energy supply. The local people will be employed to take off the operation and maintenance of the power system as well as to manage the collection of revenues from each household, which may be used for maintaining the sustainability of the system.

On top of this, subsidy possibilities – granted by ministry of power – should greatly increase the economic attractiveness of similar electricity production applications.

Appendix. Simple sizing calculations

1. Micro hydro sizing:

$$P_{\mathrm{MHG}} = \frac{9.81 \times Q \times \rho \times h}{1000} \ \mathrm{kW}$$

2. Biogas generator sizing:

$$P_{\mathrm{BGG}} = \frac{\text{Total biogas generated}\,(m^3/day) \times \mathrm{CV_{BG}} \times \eta_{\mathrm{CBG}}}{860 \times (\text{Operating hours/day})} \ \mathrm{kW}$$

3. Biomass generator sizing:

$$P_{\mathrm{BMG}} = \frac{\text{Total fuelwood available}(\mathrm{Ton/yr}) \times 1000 \times \mathrm{CV_{BM}} \times \eta_{\mathrm{CBM}}}{365 \times 860 \times (\text{Operating hours/day})} \ \mathrm{kW}$$

4. Photovoltaic generator sizing:

$$A = \frac{\left[\text{PVG Direct Load} + \frac{\text{Battery Load}}{\eta_{CC}\times\eta_{CHG}\times\eta_{DCHG}}\right] - TDE_{\text{Surplus}} \times \eta_{INV}}{H \times \eta_{PVG} \times \eta_{INV}} \text{ m}^2$$

5. Diesel generator sizing:

$$P_{DEG} = \frac{\text{TDE}_{\text{Netload}} - (\text{TDE}_{\text{Surplus}} \times \eta_{INV})}{\text{Operating hours} \times \text{Loading}(\%)} \text{ kW}$$

6. Battery sizing:

$$P_{BATT} = \frac{(\text{TDE}_{\text{Surplus}} \times \eta_{CHG}) + \text{Max surplus energy from DEG}}{\text{DOD} \times \eta_{CHG} \times \eta_{INV}} \text{ kWh}$$

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